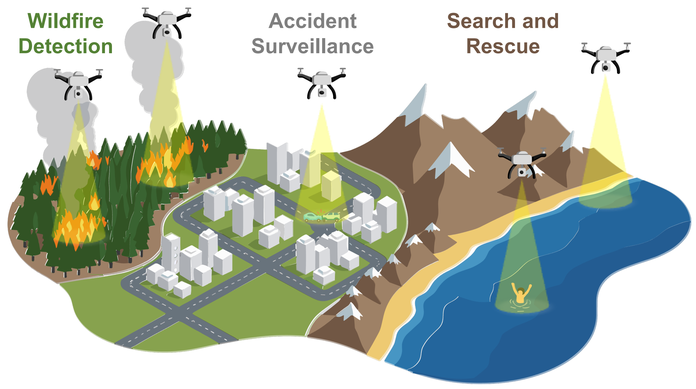
Learning Collision-Free Dispatch Policies for Route-Wise Decision-Dependent Anomaly Detection

Xin Xia[a], Chen Zhang[b], Kaibo Liu[a]

[a]Department of Industrial and Systems Engineering, University of Wisconsin-Madison; [b]Department of Industrial Engineering, Tsinghua University

Motivation & Problem

Real-world Applications

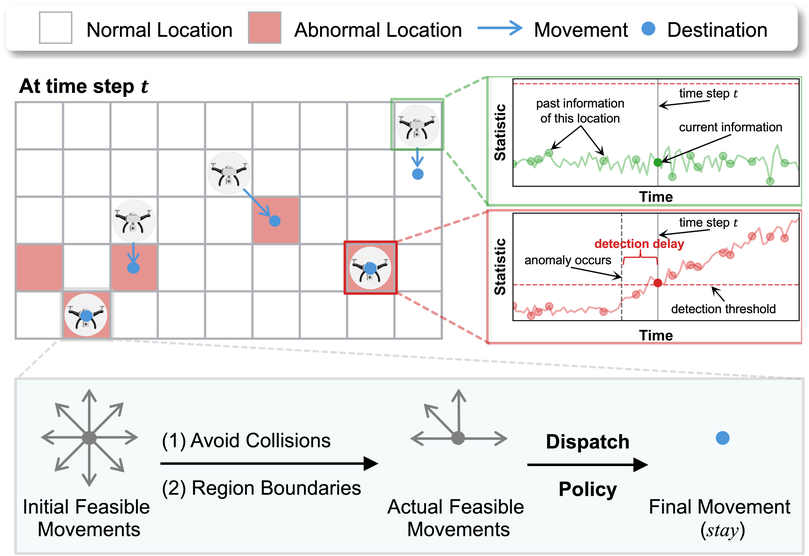


Goal: Quickest Detection under False-alarm Control

$$\min \ ARL_1(\pi, H) \quad \text{s.t.} \ ARL_0(\pi, H) \geq \alpha$$

ARL_0: mean time to false alarm
ARL_1: mean post-change detection delay

Sensing, Dispatch, and Collision-Aware Moves



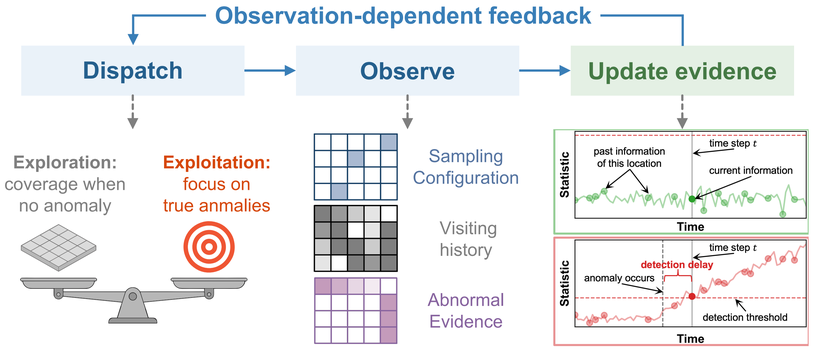
Research Gap

Track	False-alarm control	Quickest detection	Scalable policy	Multi-UAV coordination	Collision-free moves
Classic MSPC	✓	✓	△	✗	✗
Route-wise sampling	△	✓	✗	△	✗
Informative routing	✗	✗	✓	✓	✗
Safe coordination	✗	✗	△	✓	✓
Ours	✓	✓	✓	✓	✓

✓ addressed △ partial support ✗ not addressed

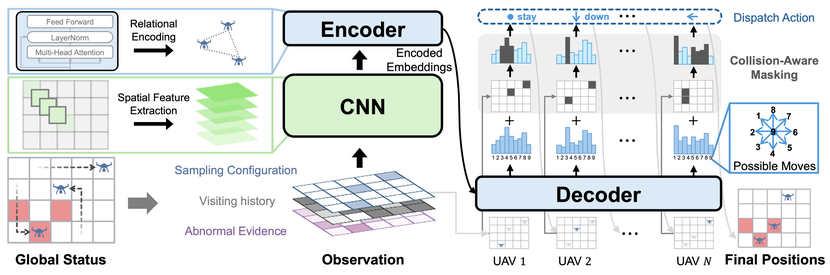
DispatchPPO: Detection-Aware Dispatch

Formulation: POMDP with decision-dependent observations



DispatchNet: samples collision-free joint moves in O(K+N)

Encoder: spatial feature + inter-UAV interaction → **Decoder:** reserve earlier UAVs' destinations for later ones



Proposition: Every joint move is collision-free and decoded in O(K+N) time.

Two-Stage Learning and Execution

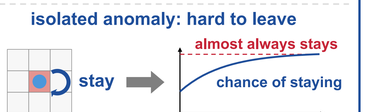
Stage I: Offline Learning
Learned Policy π_θ:
Simulated environments → PPO with proxy rewards

Stage II: Online Execution
Executed policy π^{exec}:
$\pi^{\text{exec}} = (1 - \epsilon_{\text{IC}})\pi_\theta + \epsilon_{\text{IC}}\text{Uniform Exploration}$
stay bias + ε-mixture

Theorem 1 (Persistent Coverage)
Every location keeps being visited, and none is left unobserved.



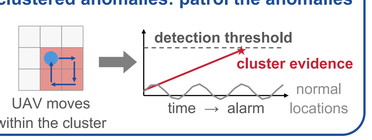
Theorem 2 (Evidence-Based Focusing)
An isolated anomaly, once revisited, becomes hard to leave.



Theorem 3 (Detection-Delay Bound)
Mean detection delay is bounded. More observations shrink the bound.



Corollary (Clustered Anomalies)
Clustered anomalies are still detected; UAVs may move within the cluster.

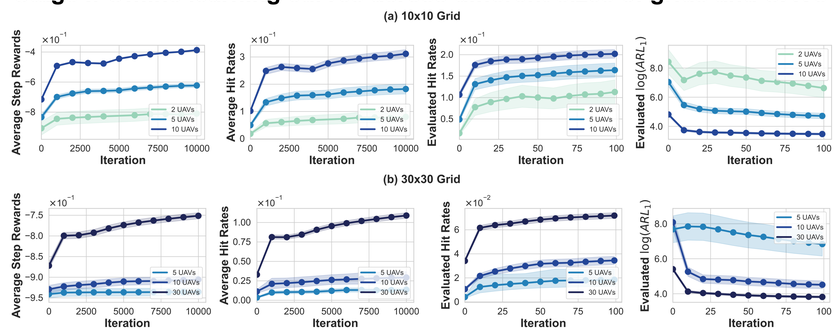


Experiments

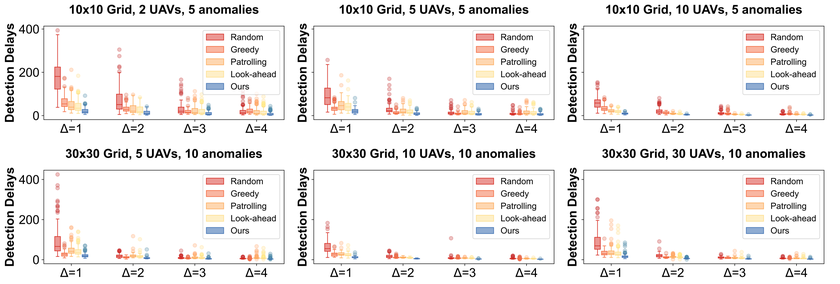
Environment	5×5 to 30×30 grids; 1–30 UAVs; 2–10 anomalies
Anomaly	mean shift Δ ∈ {1, 2, 3, 4} at τ ~ U[5, 50]
Metric	ARL_1 at ARL_0 = 370; 1,000 runs per setting

Simulations

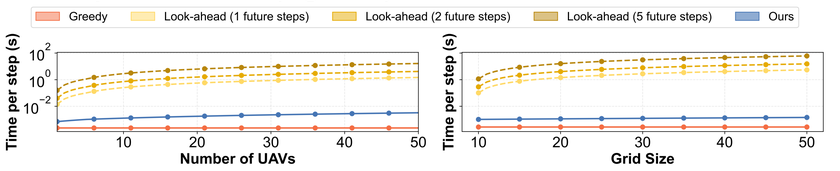
Stage I: Offline learning curves across different sizes of grids and UAVs



Stage II: Detection delay under false-alarm control

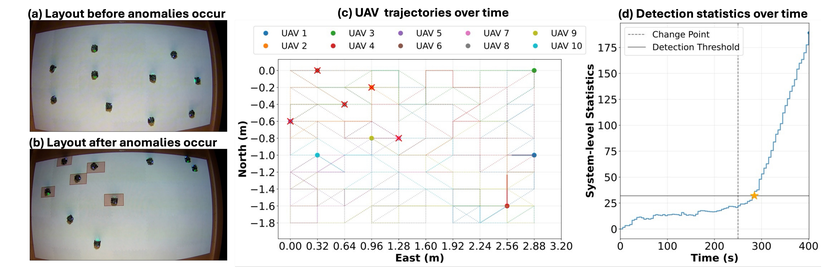


Per-step decision time (log scale)



Real-World Deployment

Robotarium: 10 robots, 5 anomalies from t = 250 s



Cherrywood wildfire (2021): 2 UAVs, 10×10 grid from satellite frames

